CS6208: Solving tiny_hana and testing CFR improvements

1. Problem, representation, and required implementation

The objective is an approximate Nash equilibrium of one round of tiny_hana, a two-player zero-sum game with hidden cards. The required game starts with neutral favors and chooses the starting player uniformly. A win pays +1, a loss -1, and an unresolved round 0. The supplied engine and treeplex representation were retained unchanged.

Task 1: beliefs. A depth-first traversal collects concrete states consistent with the target observation history. Each state receives the product of chance probabilities and opponent action probabilities, followed by normalization. The target player's own reach factor cancels under perfect recall. Inconsistent observation prefixes and zero-probability branches are pruned. Every executed action is undone in a finally block, including when an opponent raises an exception. An unreachable target returns an empty dictionary.

Task 2: best responses. Let x and y be the two realization plans and A the chance-weighted payoff matrix. Fixing y gives Player 1 reward Ay; fixing x gives Player 2 reward $-A^{T}x$. A backward treeplex pass selects the largest action value at each information set and adds it to its parent sequence. Values from multiple child information sets are added. The returned deterministic policy can be converted to sequence form.

upper = $u(BR_1(y), y)$, lower = $u(x, BR_2(x))$
SPR(x, y) = upper - lower.

This is the sum of the two unilateral improvement opportunities, without the factor 1/2 sometimes used for exploitability. For feasible strategies, lower bounds the game value from below and upper bounds it from above.

Task 3: CFR. At each information set, positive cumulative regrets are normalized into an action distribution, with uniform fallback. A backward pass computes continuation values, adds action value minus expected value to local regret, and propagates the expected value to the parent. Averages are accumulated in sequence form, ensuring correct own-reach weighting when converted back to behavioral strategies [1].

The implementation batches independent information sets by own-action depth and action count, using NumPy for regret matching and sparse reward products. It removes zero payoff entries from training. These changes preserve the backward-pass equations; they introduce no information abstraction.

2. Extension: a controlled ablation of CFR improvements

The research question is whether retaining only nonnegative regret, changing the average, or changing the update order explains faster convergence on tiny_hana. The first four configurations form a 2 x 2 ablation. They use the same simultaneous update schedule, initial zero regrets, full tree, iteration budget, and diagnostics.

Configuration	Regret update / average / schedule
CFR	Signed cumulative / uniform / simultaneous
CFR-linear	Signed cumulative / linear / simultaneous
RM+	Nonnegative cumulative / uniform / simultaneous
RM+-linear	Nonnegative cumulative / linear / simultaneous
CFR+	Nonnegative cumulative / linear / alternating

The nonnegative update is $R \leftarrow \max(R + \text{instantaneous regret}, 0)$, applied after adding regret. Linear strategy averaging uses weight t at iteration t, with no averaging delay. It does not weight the regret updates and therefore is not LCFR. In the fifth configuration, Player 2 uses Player 1's newly updated strategy. This follows the alternating, regret-matching-plus design of CFR+ [2]. Its improvement over the fourth configuration tests update order conditional on those other choices.

Game conditions. All use a uniformly random starter. Neutral has no initial favors. Advantage gives Player 1 the favor of character 1. Contested gives Player 1 character 1 and Player 2 character 3. These are three single-round payoff conditions, not a solution of the multi-round game.

Protocol. Each configuration runs for 20,000 iterations in each condition, with 3 timing repeats and rotated method order. There are 14 predetermined checkpoints from 1 to 20,000 iterations. The reported residual is evaluated exactly through best responses to the average strategy; training and evaluation use no sampled trajectories. The repeated residuals coincide, so repetitions characterize timing, not statistical uncertainty in solution quality.

Verification. Thirteen tests cover analytic posterior probabilities for both players, independently enumerated conditional histories, state restoration, exhaustive pure-strategy responses on a branching toy game, a known matrix equilibrium, and exact continuation/reset behavior. All checkpoint strategies satisfy the sequence-flow constraints within 1e-9. Separate maximin and minimax linear programs provide independent value certificates for every game condition.

3. Convergence results and interaction between improvements

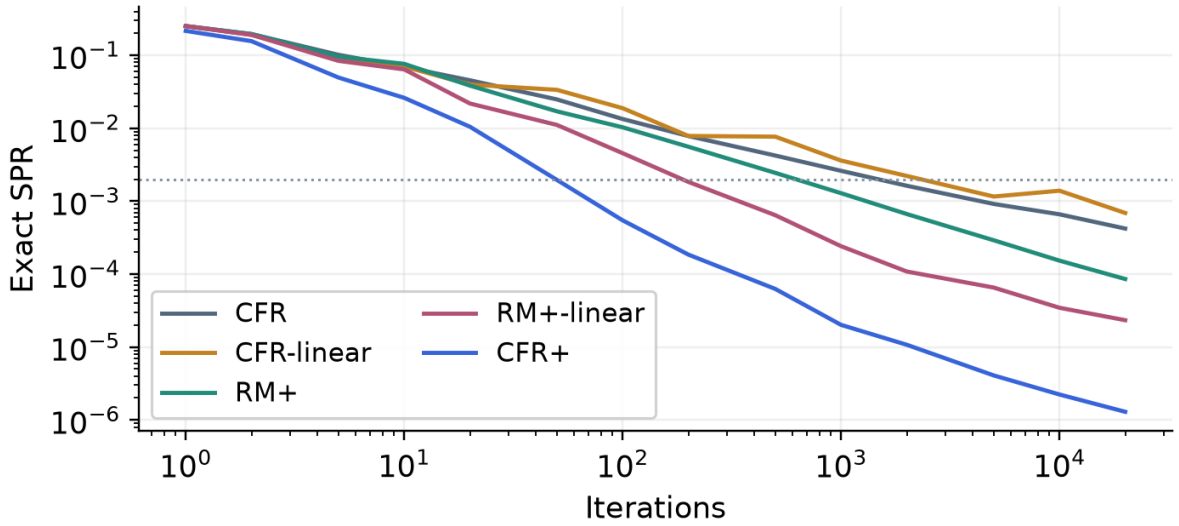


Figure 1. Neutral-game average-strategy SPR. The dotted line is the notebook example target, 0.002. Curves are exact deterministic evaluations.

Final SPR	Neutral	Advantage	Contested
CFR	0.000421	0.000439	0.000571
CFR-linear	0.000688	0.00104	0.00103
RM+	8.55e-05	0.000184	0.000241
RM+-linear	2.33e-05	8.72e-05	0.000113
CFR+	1.29e-06	1.46e-06	2.94e-06

At 20,000 iterations on Neutral, nonnegative regrets with uniform averaging reduce SPR by 4.9 times relative to vanilla CFR. Adding linear averaging to RM+ reduces it by another 3.7 times. Alternating updates then reduce it by 18.0 times. The same ranking holds in all three conditions at the final checkpoint.

Linear averaging alone is worse than uniform averaging for signed cumulative regret in every condition here. Weighting recent iterates more heavily is not sufficient when those iterates continue to oscillate. The results are consistent with an interaction between regret flooring and averaging; they do not establish a universal advantage for either choice. Intermediate curves also need not decrease monotonically.

4. Runtime, averaging, and computational cost

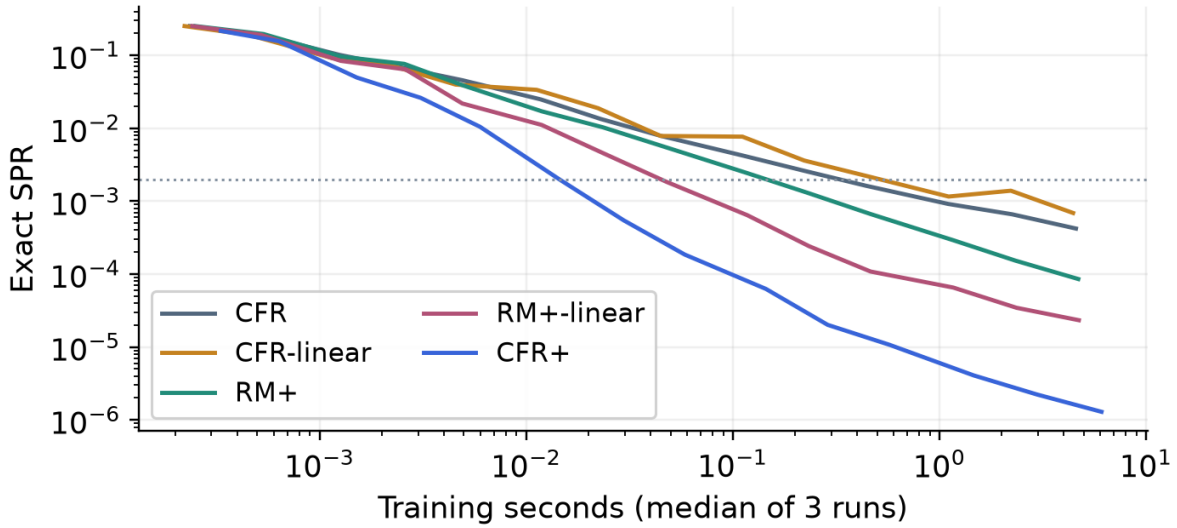


Figure 2. Neutral-game SPR against training-only time. Medians use three repetitions; exact diagnostic work is excluded from the horizontal axis.

Neutral method	Seconds: median [range]	First checked T at SPR ≤ .002
CFR	4.59 [4.49, 5.22]	2000
CFR-linear	4.44 [4.42, 5.06]	5000
RM+	4.71 [4.65, 5.01]	1000
RM+-linear	4.74 [4.67, 4.74]	200
CFR+	6.10 [5.73, 6.15]	50

Alternating CFR+ costs 1.33 times as much training time as vanilla CFR at the same iteration budget in the Neutral median. It performs an additional Player 1 strategy computation per iteration, but its lower residual also persists when compared against elapsed training time.

For vanilla CFR at the final Neutral checkpoint, the current policy pair has SPR 0.06529, while its time average has SPR 0.000421. This illustrates why the submitted artifact uses the average rather than the latest policy.

Each player has 3,672 sequences and 2,757 information sets. Neutral stores 12,832 payoff pairs, but only 2,816 are nonzero. Tree extraction is measured separately. Training time excludes payoff preparation, best-response checks, serialization, and plotting; it is not total end-to-end time. The reference LP is very fast on this small game, so these results do not argue that CFR is the fastest solver for this instance.

5. Independent certificates, limitations, and reproducibility

The independent LP uses nonnegative realization plans satisfying $E\,x = e$ and $F\,y = f$. Player 1 maximizes $f^T q$ subject to $F^T q \le A^T x$, with unrestricted q. A separate Player 2 program minimizes $e^T p$ subject to $A\,y \le E^T p$. Agreement of both optima and best-response bounds checks the value and the signs in the implementation.

LP reference	Game value (P1)	Exact BR gap
Neutral	0.00000000	6.9e-18
Advantage	0.26344671	0
Contested	0.22821429	0

The neutral value is also zero by player-swap symmetry with a uniformly random starter. The asymmetric condition checks that the LP and payoff calculations also agree when the value is nonzero.

Submitted strategy. The saved policies are the Neutral CFR+ linear time average after 20,000 iterations, with measured SPR 1.2924151e-06. They were saved and reloaded through the provided API. Validation independently checks all behavioral distributions, sequence flows, expected payoff through full engine traversal, and 100 seeded single-round games. The supplied treeplex file retains its original SHA-256.

After reloading, exact response bounds are [-6.89e-07, 6.03e-07]. Nine payoff comparisons against exhaustive engine traversal agree within 5.1e-16.

Limits. These are full-tree experiments on tiny_hana only. They do not establish scalability to medium_hana or optimality over repeated rounds. Three timing repetitions on one machine do not support precise claims about small speed differences. The final ranking is empirical, and the intermediate curves are not monotone. A useful next experiment would vary the averaging delay under a fixed time budget.

Reproduce. Run the tests, run_experiments.py, validate_submission.py, and build_report.py as documented in task4/README.md. Raw checkpoint data, all timing repeats, LP certificates, and validation results are included. The experiment environment is Python 3.13, NumPy 2.5.4, macOS 15.6.1 on arm64. Core graded code depends only on NumPy; SciPy and plotting/report libraries are optional experiment tools.

References

[1] Zinkevich, Johanson, Bowling, and Piccione (2007). *Regret Minimization in Games with Incomplete Information.* NIPS 20. Original paper.

[2] Tammelin (2014). *Solving Large Imperfect Information Games Using CFR+*. arXiv:1407.5042. Original report.